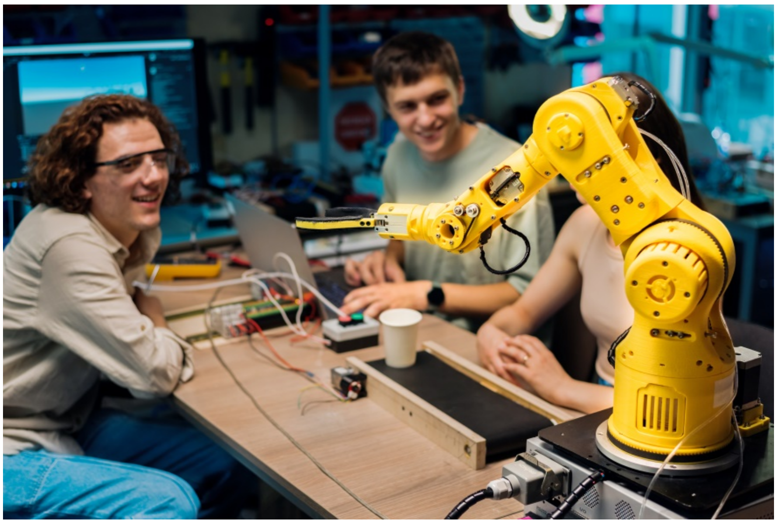
UNIT 1
ROBOTICS



Connect to the topic

1. Look at the photo. What kind of robot do you see?
2. Who might be the people in the photo?
3. What industries could benefit from using this type of robot?
4. Why do you think robotics is becoming an important field today?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video about humanoid robots and discuss the questions below.

1. What is Neo, and what tasks can it perform?
2. How can Neo be operated, according to the CEO of 1X?
3. What kind of price range is Neo expected to fall into?
4. What does Elon Musk say about the future of humanoid robots?
5. What challenges do you think engineers face when creating humanoid robots like Neo?
6. When were the words “robot” and “robotics” first used?
7. What country can be called the world´s predominant robot manufacturing country? Why?
8. What are famous robots from movies or books?

UNIT 1.1 THE WORLD OF ROBOTICS

WATCHING

Task 1. COMMUNICATE. Work in pairs. Discuss the questions below.

1. What do you know about Greek mythology?
2. What do you know about Hephaestus, King Minos, Medea, Golden Fleece, Jason and the Argonauts?
3. How can Greek myths be connected with robotics?

Task 2. EXPLORE THE WORDS. The sentences below will help you learn new words from the video. Choose the meaning of each word in bold.

1. The programmer found an **ingenious** way to fix the bug.
   a) innovative b) stupid c) artistic
2. A strong password is a **defense** against hackers.
   a) position b) protection c) barrier
3. Antivirus software helps scare off **cybercriminals**.
   a) strangers b) guests c) aggressors
4. To **forge** a blade of a knife takes great skill.
   a) to shape metal b) to produce metal c) to reuse metal
5. IT specialists are **guardians** of important data.
   a) protectors b) bodyguards c) warriors
6. The robot is strong enough to lift a **boulder** in the test.
   a) hill b) mountain c) rock
7. He got a good **bargain** on a new laptop during the sale.
   a) deal b) promise c) idea
8. Some people say AI could become **immortal** by storing its memory forever.
   a) dead b) living forever c) spoilt
9. The company created a **blueprint** for a new robot.
   a) picture b) background c) plan
10. The system crash filled the engineers with **despair**.
    a) hopelessness b) comfort c) happiness

Task 3. WATCH FOR MAIN IDEAS. Scan the QR code and watch the video "The Greek myth of Talos – the first robot" and choose the best answers to complete the statement. More than one answer is possible.

The story of Talos, first mentioned around 700 B.C. ...

A. uncovers the idea of artificial intelligence.
B. tells about the danger of artificial intelligence.
C. offers one of the earliest conceptions of a robot.
D. hints at extraterrestrial origin of Greek mythology.

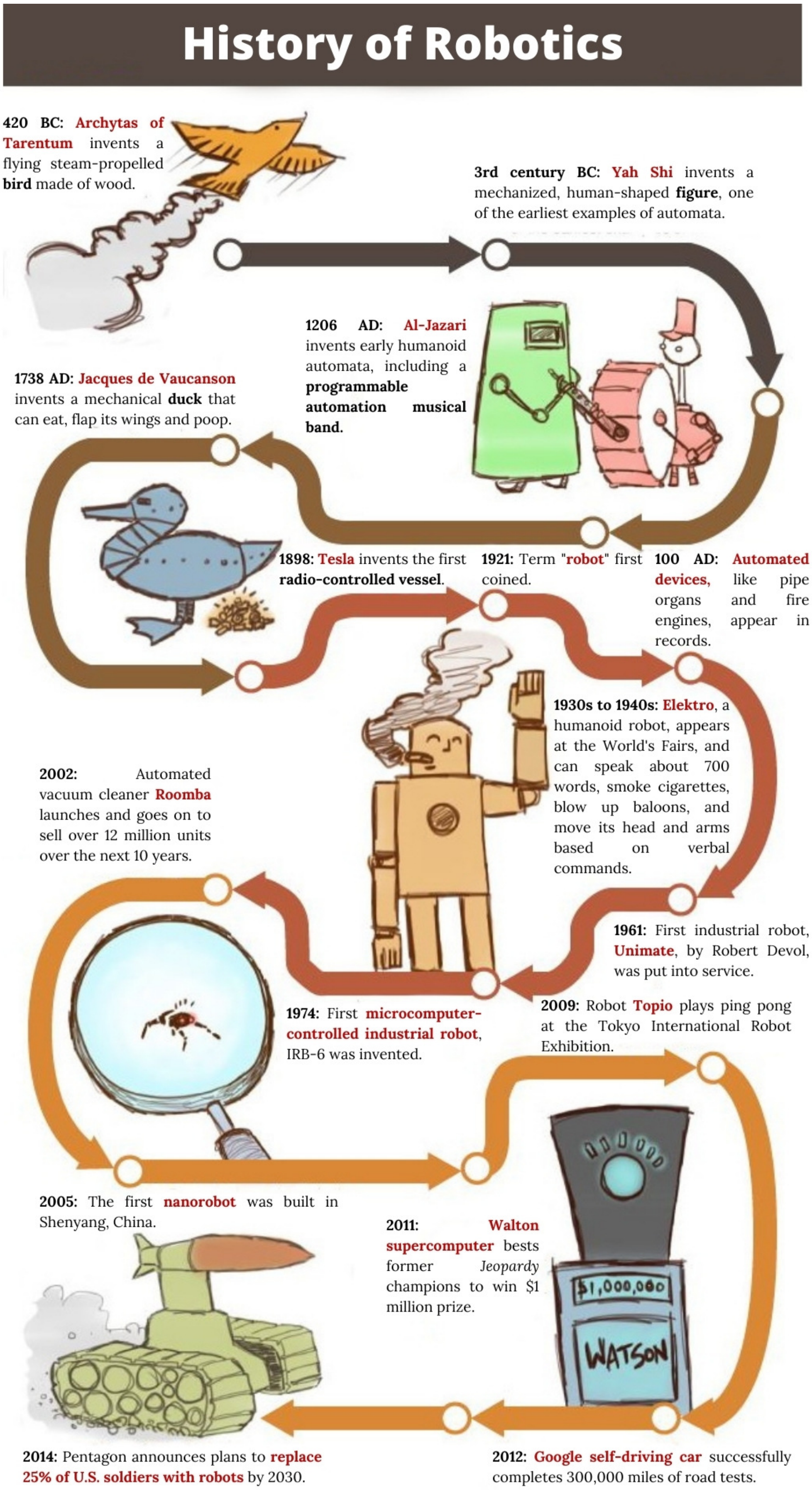
Task 4. WATCH FOR DETAILS. Watch the video again and answer the questions.

1. According to Greek myths, Hephaestus …
   A. was a god of defense.
   B. created a defense robot.
   C. destroyed a bronze automaton.
2. The bronze guardian searched for intruders …
   A. once a day.
   B. twice a day.
   C. three times a day.

3. When Talos saw the approaching ships, …
   A. his metal body heated and exploded.
   B. he threw the boulder on them.
   C. he let them leave after a talk.
4. Medea offered a bargain …
   A. to make Talos immortal in exchange for removing a bolt.
   B. to make Talos free in exchange for giving him a new bolt.
   C. to make Talos stronger in exchange for their escape.
5. Ancient Greek engineers …
   A. wrote science fiction about robotic servants and flying birds.
   B. created robotic servants and flying birds.
   C. dreamed about automatons including robotic servants and flying birds.
6. The myth about Talos cares more about …
   A. robotic brains.
   B. robotic mechanism.
   C. robotic heart.

READING AND VOCABULARY

Task 5. THINK CRITICALLY. Work in small groups to read and interpret an infographic about the history of robotics. Scan the QR code to enlarge the picture. Discuss the questions that follow.



(adapted from https://classesandcareers.com/)

1. What materials were used for the first robotic bird? How did it fly?
2. When did the term “robot” appear?
3. What humanoid robot could speak?
4. Who invented the first industrial robot?
5. What robot did Tesla invent?
6. What are Pentagon plans for the year 2030?
7. Where was the first nanorobot built?
8. Was the first vacuum cleaner robot successful?
9. What sport could a Japanese robot do at the exhibition in 2009?
10. What robotic technology did Google introduce?

Task 6. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

Robot Components

External Parts	
frame	The body or skeleton that holds the robot together.
chassis	The main supporting base of a robot, housing essential components.
gripper	A hand-like part that grabs objects.
effector	A part of the robot (e.g., gripper, claw, etc.) that interacts with the environment.
joint	A rotating connection that allows movement in elbows or knees.
robotic arm	A mechanical arm designed for manipulation and precision tasks.
tracks	Continuous belts (like on tanks) used for robots needing better stability.
housing	The outer cover that protects the robot’s insides.
speaker	An output device that allows robots to produce sound or speech.
antennae	A component used for communication and signal reception.
Internal Systems	
sensor	A part that detects things like light, sound, or walls.
motor	A part that powers movement, like spinning wheels.
processor (CPU)	The “brain” that thinks and controls the robot.
microcontroller	A small computer that processes information and controls robotic functions.
power supply	The energy source for the robot (e.g., battery, wired connection).
circuit	Wires that connect parts and carry signals or power.
actuator	A mechanical device that moves or controls parts of the robot.
gear	A toothed wheel that helps parts move together.
AI module	A system that enables robots to process data and make intelligent decisions.
navigation system	A system that helps the robot move and avoid obstacles (e.g., GPS).
cooling system	A mechanism that prevents overheating in robotic components.

1. Mud and rocks stuck between the _____________ as the robot explored the construction site.
2. The robot adjusted its ______________ to get a better signal from the control center.
3. She oiled the ___________ so the robot’s legs wouldn’t squeak anymore.
4. The _____________ clicked, lifting the robot’s arm to wave hello.
5. The robot stopped short when its ___________ spotted a chair in the way.
6. The _____________ recognized the voice command and responded immediately.
7. The robot collapsed, but its strong ____________ kept the internal parts safe.
8. The _______________ took a toy off the floor before the cat got it.

9. The robot's ______________ played 'Good morning!' to start the day.
10. Without its_____________, the robot stayed still.
11. A broken ____________ made the robot's lights go dark.
12. She fixed the wheels to the _____________ so the robot rolled easily.

Task 7. READ FOR DETAILS. Read the text and answer the questions.

How Robots Work

On the most basic level, human beings are made up of five major components: a body structure, a muscle system to move the body structure, a sensory system that receives information about the body and the surrounding environment, a power source to activate the muscles and sensors, a brain system that processes sensory information and tells the muscles what to do. Robots share these same components. A typical robot has a movable physical structure, a motor, a sensor system, a power supply, and a computer "brain" controlling these elements. Essentially, robots are human-made versions of animal life, designed to replicate movement and behavior.

Most robots have movable bodies. Some only have motorized wheels, while others have multiple jointed segments made of metal or plastic. Like human bones, the individual segments are connected together with joints. Robots move with some sort of actuators. Some robots use electric motors and solenoids as actuators; some use hydraulic systems, or pneumatic systems. Many robots may use a combination of all these actuator types. To function, they require a power source – typically batteries or plug into the wall. Hydraulic robots need a pump to pressurize the hydraulic fluid, while pneumatic robots rely on compressed air. Actuators are all wired to electrical circuits, which control motors and solenoids directly or regulate hydraulic and pneumatic systems via electrical valves. For example, to move a hydraulic leg, the robot's controller would open a valve directing fluid into a piston, extending the leg forward. Typically, in order to move their segments in two directions, robots use pistons that can push both ways.

Not all robots have sensory systems, and few can see, hear, smell, or taste. The most common robotic sense is the sense of movement – the robot's ability to monitor its own motion through navigation systems. One way to do this is to use a laser on the bottom of the robot to illuminate the floor while a camera measures the distance and speed traveled. This is the same basic system used in computer mice. Roomba vacuums use infrared light to detect objects in their path and photoelectric cells measure changes in light.

The term "robot" originates from the Czech word "robota," meaning "forced labor." This describes the majority of robots fairly well. Most robots in the world are designed for repetitive, difficult, or dangerous manufacturing work. One of the most common industrial robots is the robotic arm. Typically, it consists of seven metal segments connected by six joints. The computer controls the robot by rotating stepper motors. This allows to move the arm very precisely, performing the same movement over and over. Some larger robotic arms use hydraulics or pneumatics instead. The robot uses motion sensors to make sure it moves just the right amount.

An industrial robot with six joints resembles a human arm, it has the equivalent of a shoulder, elbow, and wrist. However, instead of attaching to a movable body, its base remains stationary. This setup has six degrees of freedom, a human arm, by comparison, has seven degrees of freedom. A robotic arm's purpose is to move an end effector, which varies based on application. One common end effector is a simplified version of the hand, which can grasp and carry different objects. Others include blowtorches, drills, or spray painters.

Robotic arms are relatively easy to build and program because they operate in fixed spaces. Things get a bit trickier when you send a robot out into the world. First, the mobile robot needs a working locomotion system. For smooth surfaces, wheels or tracks are ideal. For rough terrain, legs offer adaptability. Studying legged robots also aids biological research, as engineers mimic animal movement. Engineers often draw inspiration from animals. Six-legged robots mimic insects to have exceptionally good balance, and they adapt well to a wide variety of terrain. Four-legged robots, like Boston Dynamics' Spot, look like dogs and take on dangerous jobs such as construction site inspections. Two-legged robots, such as Atlas, demonstrate advanced balance and agility. Mobile robots vary in control systems. Some are remote-controlled via wires, radio signals, or infrared. Others operate semi-autonomously, when the operator directs the robot to go to a certain spot and the robots finds its own way.

(adapted from https://science.howstuffworks.com/robot.htm)

1. What are the main parts of a robot?
2. What connects a robot's movable segments, like bones in a human body?
3. How do robots move their body parts?
4. How does a hydraulic robot get power to move its leg?
5. What controls everything attached to a robot's circuits?
6. How does a navigation system help a robot know its own movement?
7. What are the types of end effectors mentioned for robotic arms?
8. How many joints does a typical robotic arm have?
9. Why do some robots have legs instead of wheels?
10. What kind of robots are inspired by insects?
11. How do mobile robots find their way?

Task 8. WORK WITH WORDS. Look at the highlighted words in the text and match them with their definitions.

1. **solenoid**	a) a part that pushes air or liquid to move things
2. **hydraulic**	b) a movable part inside a cylinder that slides to move robot parts.
3. **pneumatic**	c) a coil of wire that makes power to pull or push.
4. **pump**	d) using pressed air to make a robot work
5. **compressed**	e) a switch that controls air or liquid in a robot
6. **valve**	f) how free a robot is to turn or bend
7. **piston**	g) using liquid to push robot parts
8. **stationary**	h) pressed tight, like air in a tank
9. **degrees of freedom**	i) staying in one place, not moving
10. **locomotion**	j) moving from place to place, like walking

Task 9. WORK WITH WORDS. Fill in the gaps with the words from the box below. There are two extra words you do not need to use.

The Anatomy of a Robot

housing ▪ sensors ▪ piston ▪ compressed ▪ chassis ▪ solenoid ▪ pneumatic ▪ locomotion ▪ frame ▪ gripper ▪ cooling system ▪ degrees of freedom ▪

Robots are marvels of engineering, built with precision to perform complex tasks. At their core is a durable1) _____________, which serves as the backbone, supporting all other components. The 2) __________ provides stability, especially for mobile robots, while the 3) __________ encases sensitive electronics, protecting them from environmental damage.

Many robots rely on 4) __________ systems, using 5) ___________air to power actuators that drive movement. These systems enable precise control, allowing robotic arms to achieve multiple 6) __________ – twisting, bending, and reaching with remarkable flexibility.

At the end of the arm, a 7) ____________ acts as the robot's hand, designed to grasp, lift, or manipulate objects. They may include 8) ____________ to adjust their grip strength to ensure that delicate items are handled safely.

To maintain optimal performance, robots often feature a 9) ___________, preventing overheating during extended operation. This is especially critical for robots performing high-intensity tasks, such as welding or assembly.

From stationary arms to mobile robots with advanced 10) __________ systems, these machines are transforming industries, combining robust design with cutting-edge technology to handle tasks once thought impossible.

WATCHING

Task 10. WATCH FOR MAIN IDEA. Scan the QR code and watch the video "G1 Robot". Choose the best summary of the video.

A. The G1 robot is a humanoid developed for personal and household use. It is designed to assist with daily tasks and interact with users through voice commands. It features advanced AI for communication and can recognize human emotions.

B. The G1 robot is a humanoid developed by Unitree. It is designed for research and development rather than everyday consumer use. The robot has been improved to enhance its performance and adaptability in various environments.

C. The G1 robot is an industrial machine built by Unitree mainly for large-scale manufacturing. It is used in factories for assembling products and handling materials. It requires minimal supervision and can operate independently in production lines.

Task 11. WATCH FOR DETAILS. Watch the video again. Choose the correct answers to the questions.

1. What is the G1 robot known for?
   A. Being the tallest humanoid robot
   B. Having smooth and agile movement
   C. Being the cheapest robot on the market
   D. Running on solar power
2. How does the G1 compare to Tesla's Optimus robot?
   A. It is taller and heavier
   B. It is shorter and lighter
   C. It is faster but heavier
   D. It has no major differences
3. What technology helps the G1 navigate its surroundings?
   A. GPS tracking system
   B. AI voice recognition
   C. 3D LiDAR and depth camera
   D. Infrared sensors
4. How long can the G1 run on a single charge?
   A. 30 minutes
   B. 1 hour
   C. 2 hours
   D. 5 hours
5. What makes the G1's battery system more convenient?
   A. It has a quick-release system for easy swapping
   B. It can charge wirelessly
   C. It lasts longer than any other robot battery
   D. It does not need charging

6. How much does the G1 robot cost?
   A. $10,000
   B. $16,000
   C. $25,000
   D. $30,000
7. Who is the G1 robot mainly designed for?
   A. Developers and researchers
   B. Ordinary consumers
   C. Schools for children
   D. Factory workers

SPEAKING

Task 12. COLLABORATE. Work in pairs. Draw a robot on a separate sheet of paper, including parts from the vocabulary list (e.g., pistons, sensors, wheels, joints). Add unique features like extra arms or cameras. Keep your drawing hidden from a partner. Describe your robot to your partner while he or she draws based on your description. Your partner can ask questions but cannot see the original. Once finished, compare drawings. Discuss similarities, differences, and any misunderstandings. Then switch roles and repeat.

Task 13. COLLABORATE. Work in pairs or small groups. Choose a job for your robot (e.g., firefighter, doctor, factory worker, space explorer, security guard, farmer, chef, delivery, etc.). Discuss its design, and then present it to the class.

Ideal Robot for a Job

Size & shape – Is it tall, small, humanoid, or machine-like?
Materials – What is it made of (metal, plastic, special alloys)?
Arms/legs – How many does it need? Does it have wheels or tracks instead?
Sensors – What does it need to see, hear, or detect (cameras, heat sensors, motion detectors)?
Power source – Does it run on electricity, solar power, batteries, or something else?
Explain:

- What your robot looks like.
- How it works.
- Why it is useful for the job.

Task 14. COLLABORATE. Divide into two teams to have a debate about a robot design.
Some say robots should look like humans, while others think they should be more like machines. What do you think? Human-like robots can feel friendly and work in homes, but they might be hard to build. Machine-like robots can be strong and simple, but they might not fit everywhere. What's your opinion?

Should robots look like humans or like machines?

Human-like	Machine-like
1. Human-like robots can be friendly and helpful in homes, using grippers and joints to do tasks like us. 2.	1. Machine-like robots are stronger and simpler to build, with tracks and a tough chassis for hard jobs. 2.

Task 15. COMMUNICATE. Role play the situations below.

Robot Repair Shop

Student A: You own a robot that has a problem. Choose two or three issues from the list below and explain them to the repair technician. Answer any questions about the problem.

- The robot won't turn on, even though it is charged.
- One of its arms or legs doesn't move properly.
- It doesn't detect obstacles and keeps bumping into things.
- It doesn't respond to voice commands.
- It overheats and shuts down after a few minutes.
- The battery drains too quickly.
- It makes strange noises when it moves.

Student B: You are a repair technician. Ask questions to understand the issue and suggest possible solutions.

Ordering a Custom Robot

Student A: You are a customer visiting a robotics company to order a robot for a specific job. Choose the type of job the robot should do (e.g., cleaning, warehouse assistance, medical assistance, underwater exploration) and explain the details to the engineer. Ask about the robot's features and how it works.

Student B: You are an engineer at the robotics company. Ask questions to understand the customer's needs and suggest features based on the job the robot will do. Describe how the robot will perform its tasks and explain its capabilities.

Task 16. COLLABORATE. Work in small groups. Your team is part of the robot investigation unit, you are trying to find the missing robot and understand what has happened. You received a case file with 3 main clues: sensor logs from the last hour, GPS trail showing unexpected movement, command history from the control system. Some experts think it was a software error. Others believe the robot was hacked or reprogrammed. A few even say the robot may have developed independent behavior... Discuss what might have happened:

- Did the robot malfunction?
- Was it stolen or hacked?
- Could it be a sign of rogue AI?

Choose one theory your group agrees on. Prepare a short report or presentation explaining: what happened, why you think so (evidence from clues), what should be done next.

Clue 1 Sensor Logs (Last Hour of Activity)	Clue 2 GPS Trail	Clue 3 Command History
13:58 – Package secured 14:02 – Proximity sensors clear 14:05 – Sudden obstacle detected (no object found) 14:08 – Emergency stop activated for 2 seconds 14:09 – Resume movement 14:13 – Temperature sensor spike (+10°C in 2 seconds) 14:17 – Visual sensor offline 14:19 – Motion detected without command 14:24 – All sensors normal 14:31 – Battery dropped from 78% to 12% 14:47 – No signal	Expected Route: Zone A → Zone C → Zone D Actual Movement: 14:00: Zone A 14:06: Entered Zone C 14:12: Stopped in Zone C 14:14: Turned toward loading dock (unauthorized area) 14:20: Brief location jump – coordinates not found 14:22: Back in Zone C, near old server room (decommissioned) 14:30: Moved 3 meters into wall (unreal location) 14:47: Signal lost	Last 10 Commands: 13:56 – Start delivery route 13:58 – Confirm load 14:00 – Begin movement 14:05 – Pause: object detected 14:09 – Resume 14:12 – Route deviation requested (unknown user ID) □ 14:14 – Command override accepted 14:16 – Diagnostic report blocked 14:17 – Visual feed disabled 14:19 – Command loop started: “Free path” (repeats)

Task 17. ANALYZE. Read the case study and discuss possible solutions to the problem.

A university campus used autonomous delivery robots to transport food and packages between buildings. The robots relied on a combination of GPS, cameras, and infrared sensors to detect obstacles. One day, several robots stopped mid-route during heavy fog. The infrared sensors failed to distinguish between real barriers and mist, and the GPS signal weakened due to atmospheric interference. As a result, some orders were delayed or canceled.

How can the design of robotic sensor systems be improved to ensure reliable navigation in different weather conditions?

LANGUAGE FOCUS

Task 18. STUDY AND ANALYZE. Look at the rule about reported speech in statements, study in what situations they are used.

REPORTED SPEECH IN STATEMENTS

Direct Speech	Reported Speech
Present Simple ‘I **need** a new robot,’ said Molly.	**Past Simple** Molly said she **needed** a new robot.
Present Continuous 'I’**m programming** a robot assistant,' said Alex.	**Past Continuous** Alex said he **was programming** a robot assistant.
Present Perfect 'I’**ve built** an AI-powered drone,' said Lisa.	**Past Perfect** Lisa said she **had built** an AI-powered drone.
Present Perfect Continuous 'I’**ve been working** on a robotic arm for months,' said Tom.	**Past Perfect Continuous** Tom said he **had been working** on a robotic arm for months.
Past Simple 'I **tested** the new robot yesterday,' said the developer.	**Past Perfect** The developer said he **had tested** the new robot yesterday.
Past Continuous 'I **was designing** a robot for medical use,' said	**Past Perfect Continuous** Sarah said she **had been designing** a robot

Sarah.	for medical use.
Past Perfect 'I **had installed** the latest software before the robot started working,' said Mike.	**Past Perfect** (no tense change) Mike said he **had installed** the latest software before the robot started working.
Past Perfect Continuous 'I **had been training** the AI model for weeks before the competition,' said Jack.	**Past Perfect Continuous** (no tense change) Jack said he **had been training** the AI model for weeks before the competition.
Am/is/are going to 'I'**m going to** test the robot's sensors tomorrow,' said Emma.	**Was/were going to** Emma said she **was going to** test the robot's sensors tomorrow.
Will 'The robot **will** improve efficiency,' said the scientist.	**Would** The scientist said the robot **would** improve efficiency.
Can 'This robot **can** recognize human speech,' said the engineer.	**Could** The engineer said the robot **could** recognize human speech.
Must / have to 'We **must** update the robot's firmware,' said the technician.	**Had to** The technician said they **had to** update the robot's firmware.
May 'The robot **may** need a software upgrade,' said the developer.	**Might** The developer said the robot **might** need a software upgrade.

Watch out!

We only make tense changes if the reporting verb (say, etc.) is in the past.

Jack says he is going to study robotics.

~~Jack says he was going to study robotics.~~

We do not need to make any changes to the verb tense or modal when we are reporting a scientific fact or when something is still true.

'Industrial robots work 24/7 without getting tired,' said the engineer. – The engineer said industrial robots work 24/7 without getting tired.

Pronoun and Determiner Changes

Direct Speech	Reported Speech
My 'I've updated **my** AI system,' said Sarah.	**His/her** Sarah said she had updated **her** AI system.
This 'I designed **this** robot,' said Alex.	**The/that** Alex said he had designed **the/that** robot.
These 'I tested **these** sensors,' said the engineer.	**Those** The engineer said he had tested **the/those** sensors.

Time and Place Changes

Direct Speech	Reported Speech
Here 'The robot is charging **here**,' said the engineer.	**There** The engineer said the robot was charging **there**.
Now / at the moment 'The AI is processing data **at the moment**,' said the developer.	**Then / at that moment** The developer said the AI was processing data **then / at that moment**.

Tomorrow 'I will test the robot **tomorrow**,' said Alex.	**The next / following day** Alex said he would test the robot **the next / following day**.
Tonight 'The team is presenting the new AI model **tonight**,' said the manager.	**That night** The manager said the team was presenting the new AI model **that night**.
Next week / month / year 'We are launching the autonomous drone **next month**,' said the CEO.	**The following week / month / year** The CEO said they were launching the autonomous drone **the following month**.
Yesterday 'I fixed a bug in the robot's code **yesterday**,' said Sam.	**The day before / the previous day** Sam said he had fixed a bug in the robot's code **the day before / the previous day**.
Last week / month / year 'The robot prototype was completed **last week**,' said the project lead.	**The week / month / year before / the previous week / month / year** The project lead said the robot prototype had been completed **the week before / the previous week**.
Ago 'I updated the machine learning algorithm two days **ago**,' said the engineer.	**Before / previously** The engineer said he had updated the machine learning algorithm two days **before / previously**.

Task 19. PRACTICE. Choose the correct variant.

1. 'This robot can recognize human emotions,' said Dr. Miller.
   Dr. Miller said the robot _____ recognize human emotions.
   A. can
   B. could
   C. must
2. 'I'm working on a new robotic arm prototype,' said Sofia.
   Sofia said she _____ on a new robotic arm prototype.
   A. was working
   B. is working
   C. has worked
3. 'The robot has completed the task,' said Alex.
   Alex said the robot _____ the task.
   A. has completed
   B. had completed
   C. will complete
4. 'The delivery drone was flying too low,' said the engineer.
   The engineer said the delivery drone _____ too low.
   A. flew
   B. was flying
   C. had been flying
5. 'You must update the software regularly,' the instructor said.
   The instructor said we _____ update the software regularly.
   A. had to
   B. must
   C. could
6. 'The robot may need a software upgrade,' said Jake.
   Jake said the robot _____ need a software upgrade.

A. must
B. could
C. might

7. 'At noon, I was testing the robot's sensors,' said Mia.
   Mia said she _____ the robot's sensors at noon.
   A. was testing
   B. had tested
   C. had been testing
8. 'They will build a humanoid robot,' said the scientist.
   The scientist said they _____ a humanoid robot.
   A. built
   B. would build
   C. had built

Task 20. PRACTICE. Complete using the words and phrases in the boxes.

Words: his • their • them • there	Phrases: before • that night • the day before • the next day

1. 'They are delivering the robot tomorrow,' said Emma.
   Emma said they were delivering the robot _____.
2. 'We developed the AI system two years ago,' said the engineers.
   The engineers said they had developed the AI system two years _____.
3. 'Our team is presenting the new robotic arm,' said the developers.
   The developers said _____ team was presenting the new robotic arm.
4. 'My AI assistant gave me accurate data!' said Liam.
   Liam said _____ AI assistant had given him accurate data.
5. 'I saw the delivery drones in action at the expo yesterday,' said Sarah.
   Sarah said she had seen the delivery drones _____.
6. 'Famous robotics engineers greeted us at the event,' said Noah and Emma.
   Noah and Emma said famous robotics engineers had greeted _____ at the event.
7. 'I left my prototype here,' said Mark.
   Mark said he had left his prototype _____.
8. 'I will charge the robot tonight!' said Leo.
   Leo said he would charge the robot _____.

Task 21. PRACTICE. Complete using the correct form of the verb in brackets.

1. Jack said his new robot _________ **(not / work)** properly because of a software error.
2. Anna told us that engineers _________ **(develop)** a robot with artificial intelligence recently.
3. The scientist said that the robot _________ **(detect)** movement using advanced sensors.
4. Peter said his robot _________ **(try)** to recognize voices, but it still needed improvements.
5. The technician said they _________ **(think)** of upgrading the robot's processor.
6. Our professor explains that robots _________ **(be)** essential in modern manufacturing.
7. Michael said his robotic assistant _________ **(be going)** to clean the house automatically.
8. Laura mentioned that during the presentation the robot _________ **(love)** interacting with humans, or at least, it seemed so.
9. The programmer said he _________ **(ask)** the robot to perform a task, but it failed.
10. David told us that he _________ **(bring up)** his robotic pet with smart learning algorithms the following month.

UNIT 1.2 ROBOTICS IN DAILY LIFE

LISTENING

Task 22. LISTEN FOR DETAILS. Scan the QR code and listen to "Robotics by the numbers" about some statistics in robotics. Match the sentences to the figures mentioned in the listening. There are extra figures you do not need to use.

1. The estimated value of the global robotics market in 2023.
2. The expected value of the global robotics market by 2030.
3. The number of industrial robots operating worldwide today.
4. The number of joints in a typical industrial robotic arm.
5. The cost range (in dollars) for installing an industrial robotic arm.
6. The accuracy rate of robotic-assisted surgeries.
7. The percentage of households expected to own a robot by 2030.
8. The maximum weight (in kilograms) that some autonomous delivery robots can carry.

$50,000 - $150,000 ▪ 95% ▪ 6 ▪ 4 million ▪ 80% ▪ 20 ▪ $ 71.2 billion ▪ 60% ▪ $200 billion ▪ $1 trillion ▪ 10 million ▪ $500,000 ▪ $15,000 - $150,000

Task 23. LISTEN FOR DETAILS. Listen to the report again. Decide whether the following statements are True, False, or Not Given.

1. The global robotics market is growing rapidly.	True / False / Not Given
2. Industrial robots need frequent breaks during operation.	True / False / Not Given
3. Robotic-assisted surgeries are more accurate than traditional procedures.	True / False / Not Given
4. The cost of all robotic arms is over $100,000.	True / False / Not Given
5. Some robots can recognize up to 50 different facial expressions.	True / False / Not Given
6. Autonomous delivery robots are already delivering goods worldwide.	True / False / Not Given
7. By 2030, most factories will stop using human workers completely.	True / False / Not Given
8. The impact of robotics on daily life is expected to increase.	True / False / Not Given

Task 24. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Do you think the prediction about 80% of households expected to own at least one robotic device will come true? Why or why not?
2. Why do you think there is such a big price difference for industrial robots?
3. How might increasing robot use in industries affect job opportunities for people?
4. Do you think recognizing human facial expressions by a robot is useful? Why or why not?
5. Should there be laws to limit how much robots can do? Why or why not?

WATCHING

Task 25. WATCH FOR DETAILS. Scan the QR code and watch the video "Types of Robots". Match each robot type to the correct description. One description is extra.

1. Autonomous Mobile Robots (AMRs)
2. Automated Guided Vehicles (AGVs)
3. Articulated Robots
4. Humanoids
5. Cobots
6. Hybrids

A. These robots follow fixed paths or tracks and are used for material transport in warehouses.
B. These robots work alongside humans, helping them perform tasks efficiently.
C. These robots are a mix of different robotic types, allowing them to complete complex tasks.
D. These robots mimic human form and can perform tasks like lifting objects and navigating difficult terrain.
E. These robots move independently, using sensors and AI to navigate environments without human control.
F. These robots have joints that allow for flexible movement, often used for tasks like welding or packaging.
G. These robots are only used for cleaning and disinfecting hospital supplies.

Task 26. COMMINICATE. Work in pairs. Discuss the questions below.

1. Which type of robot do you think will be the most important in the future? Why?
2. Would you feel comfortable working with a robot? What tasks would you want it to help with?
3. If you could design a hybrid robot, what functions would it have? What problems would it solve?
4. Do you trust robots to make important decisions, such as in healthcare or security? Why or why not?
5. Imagine you have a personal robot assistant. What tasks would you ask it to do?
6. Are there any tasks that robots should never do? Why?
7. What are some possible dangers of using robots in everyday life? How can we prevent them?
8. Do you think robots should have emotions? How would that change the way we use them?

READING AND VOCABULARY

Task 27. COMMUNICATE. Work in pairs. Discuss the questions below.

1. What do you know about robots being used to help elderly people?
2. What kinds of tasks do you think robots could do in elderly care?
3. Do you think robots can replace human caregivers? Why or why not?

Task 28. READ FOR DETAILS. Read the text and choose the correct answer for each question.

Robots for Elderly Care

Japan's nursing homes feature PARO, a robotic seal that dementia patients hold close, easing their isolation with a strange sort of comfort. Across the globe, robots are entering elderly care, driven by longer lives and widely spread families. These machines handle practical needs: health checks, daily tasks, even companionship. This offers seniors a chance to stay independent longer than before.

The concept isn't new. Early tools like robotic vacuums made the way, but now artificial intelligence allows more complex helpers. Some of them remind you to take a medication or carry groceries across a room. Light and durable exoskeletons support weak legs, extending mobility where it might otherwise disappear. Others monitor heart rates and oxygen levels, ready to call for help if they fall. Japan has pushed this tech for years, but adoption has been difficult – only about 1 in 10 care facilities uses it, slowed by high costs and installation challenges.

Companionship stands out as a quieter benefit. Stevie, a robot in care homes, talks with residents and helps with daily routines. Pepper, a humanoid model, keeps seniors engaged when they fall into loneliness. PARO's plush form elevates mood and relieves stress for people with memory loss. Beyond emotions, these tools increase safety with emergency alerts and health tracking, which allows problems to be identified before they worsen. Medication arrives on schedule, and interactive features – games or video calls – keep minds active and connected.

Japan's long effort shows both promise and limitations. Decades of investment have produced PARO, Pepper, and more, yet their price leaves much to be desired – a single device can cost thousands. Safety issues also raise complaints; a malfunction can disrupt care. Still, the impact shows: mobility improves, loneliness is reduced, and staff is given a freedom of action. Other countries, facing their own aging waves, study this closely, thinking about how it might fit their systems.

Looking ahead, smarter robots could refine the approach. Advanced AI will allow them to adapt to human quirks – ensuring pills are taken or helping with a coat. That could mean more years at home, safer and more comfortable, which will be a real win for dignity. But there's a catch. Care depends on human warmth – something that metal, chips and circuits can't fake. Japan's slow adoption, despite its head start, proves that integration requires more than innovation. Institutions are relying on these tools to stretch resources, and they are delivering results – Stevie's chatter, PARO's calm, and exoskeletons' strength. The numbers prove it: health care oversight is getting tighter; falls are decreasing, and routine is staying the same.

Yet the balance matters. Machines handle tasks, not feelings. Older people get such benefits as independence, security, a bit of company, but the best care blends tech with people. Robots push the boundaries of what's possible, no doubt. They're not here to replace anyone, they just help, and they do it well enough to keep watching.

(adapted from https://care.samarth.community/blog/memory-cognition/robots-for-senior-care/)

1. What is one key reason robots are used in elderly care?
   A. They allow seniors to live independently longer
   B. They are cheaper than human caregivers
   C. They replace family visits entirely
   D. They are easier to use than smartphones
2. Why are robots becoming more common in elderly care?
   A. Families are moving closer to elderly relatives
   B. Robots have completely replaced traditional caregivers
   C. Seniors live longer, and families are often far away
   D. Nursing homes prefer robots over human staff
3. Which of the following is an example of what elderly care robots can do?
   A. Monitor health and assist with daily tasks
   B. Perform surgeries in nursing homes and hospitals
   C. Help elderly people to invest money
   D. Make legal decisions for seniors
4. Why is robot adoption in elderly care slow in Japan?
   A. Government introduces restrictions on robot use
   B. Care facilities prefer traditional methods

C. High costs and installation difficulties
D. Seniors refuse to use technology

5. How do companionship robots like PARO help seniors?
   A. They track physical exercise routines
   B. They provide emotional support and reduce stress
   C. They help seniors walk
   D. They assist with household repairs
6. What is a potential risk of using robots in elderly care?
   A. They may not always function correctly
   B. They completely replace human caregivers
   C. They are too advanced for seniors to understand
   D. They make healthcare more expensive
7. What future improvement could make elderly care robots more useful?
   A. Advanced AI that better adapts to human needs
   B. Replacing all human caregivers with robots
   C. Making robots more independent from users
   D. Reducing government regulations on robotics
8. What is the main conclusion of the text?
   A. Robots are better than human caregivers
   B. Elderly people dislike robots in nursing homes
   C. The best care combines technology with human support
   D. Robots should fully replace nursing home staff

Task 29. COMMUNICATE. Work in pairs. Role play the situation below.
Student A: You are a robotics developer trying to sell an elderly care robot to a nursing home. Explain how the robot can help with daily tasks, improve safety, and provide companionship. Answer all questions about any concerns about cost, safety, or emotional impact. Try to persuade the director that this technology will benefit both the elderly and the staff.
Student B: You are the director of a nursing home. You are in doubt about using robots in elderly care due to concerns about cost, safety, and whether robots can truly replace human interaction. Ask questions to understand how the robot works and consult with the developer about potential problems before making a decision.
Useful phrases:

- This robot can assist with…
- One major concern is…
- How does the robot ensure safety?
- What happens if the robot malfunctions?
- This will improve the quality of care because…
- Can it adapt to different patients' needs?
- While robots can't replace humans, they can help by…

Task 30. WORK WITH WORDS. Complete the sentences with the correct word. There are two extra words you do not need to use.

industrial ▪ healthcare ▪ vacuum cleaner ▪ logistics ▪ humanoid ▪ agricultural ▪ underwater ▪ rescue ▪ bomb disposal ▪ surveillance ▪ autonomous tractors ▪ rover ▪

1. Robots in ______ help surgeons perform more precise operations.
2. ______ robots are used in space to explore other planets.

3. Amazon uses ______ robots in its warehouses to move packages.
4. ______ drones help farmers monitor crop health.
5. ______ robots assist in dangerous environments such as disaster zones.
6. In manufacturing, ______ robots help assemble products quickly and accurately.
7. ______ robots are designed to interact with humans and assist in daily tasks.
8. ______ robots are used in law enforcement to detect and neutralize explosives.
9. ______ robots patrol offices, shopping malls, and parking lots for security purposes.
10. ______ robots help scientists study marine life and explore deep-sea environments.

Task 31. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.

Grease the wheels – to make a process smoother or more efficient, often with tech adjustments.
*Adding a new sensor **greased the wheels** for the robot's navigation.*
Wheels within wheels – a complex system with many interconnected parts.
*The robotics lab was full of **wheels within wheels**—gears, code, and AI.*
Cog in the machine – a small but essential part of a larger system.
*As a junior coder, I felt like a **cog in the machine** of the robotics project.*
Throw a wrench in the works – to disrupt or complicate a process unexpectedly.
*A power surge **threw a wrench in the works** during the robot's test run.*
Run like clockwork – to operate smoothly and predictably.
*After debugging, the robot **ran like clockwork** on the assembly line.*
Off the rails – to go out of control or deviate from the plan.
*The robot's AI **went off the rails** and started stacking boxes upside down.*
Bolt from the blue – an unexpected event or innovation.
*The new robotic arm design was a **bolt from the blue** in the industry.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. The team was shocked when a competitor's sudden breakthrough came like a ______ at the robotics expo.
2. A glitch in the software _______ during the live demonstration, causing chaos on stage.
3. With all the motors, circuits, and programming, building the robot felt like managing _______.
4. After hours of fine-tuning, the robotic delivery system finally _______ without any errors.
5. The intern saw himself as just a ________, quietly fixing small bugs in the robot's code.
6. Adding a backup battery really ______ for the robot's performance during long shifts.
7. When the navigation system failed, the prototype went _______ and crashed into the wall.

Task 32. COLLABORATE. Work in pairs. Imagine a factory where robots are in charge, but one day, a malfunction sends everything into chaos. How do the human workers react and fix the situation? Use at least three idioms in your scenario.

Task 33. COLLABORATE. Work in small groups. Design an unusual robot for everyday life (e.g., a pet-feeding bot or a homework helper). Pitch it to the class as if you're a robotics startup. Use at least two idioms in your pitch. Vote on the best robot idea as a class.

SPEAKING

Task 34. ANALYZE. Work in pairs. Look at a list of modern robots and categorize them into the correct groups based on their function. If you are unsure about any of them, use the internet to find more information. Once you have categorized all the robots, discuss your choices with your partner and explain your reasoning.

Domestic Robots (used in households)	**Medical Robots** (used in healthcare)	**Industrial Robots** (used in factories and production)	**Exploration Robots** (used for space, ocean, or disaster site exploration)

- Roomba
- Da Vinci
- Boston Dynamics Spot
- ASIMO
- Curiosity Rover
- Robonaut 2
- Pepper
- PackBot
- KUKA
- Moxi
- Aibo
- Perseverance Rover

1. Why did you place each robot in that category?
2. Were any of the robots difficult to classify? Why?
3. Do any of the robots belong to more than one category?
4. Which of these robots do you think will have the biggest impact in the future?

Task 35. COMMUNICATE. Work in pairs. Choose one robot from the list above and argue that it is the most useful modern robot. Present three reasons to support your choice.

1. Why is this robot more beneficial than others?
2. Does this robot have any disadvantages?
3. If you could improve it, what would you change?

Task 36. COLLABORATE. Work in groups. Read robots descriptions and discuss the strengths and weaknesses of each robot. Compare the robots based on their functionality, usefulness, uniqueness. Decide which robot is the most useful and explain your choice.

Modern Domestic Robots

Roborock Saros Z70

At a glance, this device might seem like just another robot vacuum. But there's one clever feature that stands out – a robotic arm, referred to as OmniGrip, which is designed to pick up small loose items like socks or cords before the cleaning starts. This not only avoids clogs but also makes the cleaning process smoother. Its movement relies on AI to recognize and avoid obstacles such as pets or furniture. Interestingly, it doesn't stop at vacuuming – it can also mop, making it a kind of hybrid cleaner.

Samsung Ballie

Although it resembles a toy, a small, rolling sphere, Ballie is in fact a mobile home assistant. It can follow the user from room to room and operate connected smart devices, such as lights, thermostats, and TVs. Its built-in camera enables it to check on pets or monitor specific areas of the house. Among its features are task reminders, a wake-up function, and basic entertainment support. The idea is to provide hands-free assistance that moves with the user.

Tesla Optimus

Tesla's Optimus project focuses on developing a humanoid robot that could eventually support users in domestic environments. Some of the basic tasks it's designed for include carrying light items, folding clothing, and helping with simple meal preparation. According to Tesla, its long-term goal is to automate repetitive or physically demanding household work. What differentiates Optimus from many current robots is its mobility, which is modeled after human movement; this allows it to move in spaces built for people.

Amazon Astro

Astro blends home monitoring with voice-assistant functionality. Outfitted with cameras and facial recognition, it can patrol rooms, check on elderly residents, and identify familiar people. It integrates fully with Amazon's Alexa system and can carry small objects like drinks or medication from one room to another. In some setups, users also rely on Astro for video calls, as it can bring the screen to the person, rather than the other way around.

Moley Robotic Kitchen

Moley is not your average kitchen appliance. It's a system with two robotic arms that hang above the stove, moving in a way that mimics human chefs. These arms are programmed to perform tasks like chopping and stirring, and yes, even plating a finished dish. Users can choose from a set of stored recipes. While it's not something you'd see in most homes yet, it does show what full kitchen automation might look like. The cleanup feature is also built in, though how effective it is can vary depending on the task.

1. Which robot would you most like to have at home? Why?
2. Which one do you think is the most innovative?
3. Which robot could completely replace human effort for its task? Which one still needs human assistance?
4. How could each of these robots be improved?
5. If you could design your own domestic robot, what features would it have?

TASK 37. COLLABORATE. Work in small teams. Each student randomly draws 1 Robot Card and 1 Location Card. Discuss:

- Why the robot is not suitable for the location.
- How you could adapt or reprogram it to be useful there.

At the end vote for the most creative or realistic solution.

Example: Okay, so imagine you've got a surgical robot, you know, the kind they use for really precise operations in hospitals, and now it's supposed to work in a school cafeteria. Yeah… that doesn't sound like a great match at first. It's not built for food, obviously. But maybe there's a way to make it work. Like, what if you swapped out the scalpels for kitchen tools? I'm thinking knives, maybe even a cake slicer or something like that. And instead of recognizing organs or tissues, it could be reprogrammed to identify food items: sandwiches, fruit, whatever. It might actually be amazing at slicing super neat portions. Like, every piece of cake would be exactly the same size. Weird idea, but kind of cool too.

Robots Cards

Museum guide robot	Industrial welding robot	Farming robot
Robot barista	Underwater exploration robot	Bomb-disposal robot
Delivery drone	Teaching assistant robot	Warehouse robot
Cleaning robot	Traffic-control robot	Construction robot
Robotic chef	Companion robot for the elderly	Humanoid receptionist robot

Location Cards

Swimming pool	Hiking trail in the forest	Public beach
Kindergarten classroom	Art museum	Train station platform
Concert stage	Football stadium	Dentist's office
Luxury fashion boutique	Space shuttle	Hair salon
Ski resort	Camping site	Library

Useful phrases:

- This robot is clearly not designed for…
- However, we could adapt it by…
- If we reprogrammed it to…, it might actually be helpful in…
- It would be funny but surprisingly useful if…

Task 38. ANALYZE. Read the case study and discuss possible solutions to the problem.

A family recently bought a domestic robot that could do several tasks, things like vacuuming, bringing objects, or switching off appliances. It was supposed to respond to voice commands and navigate the apartment using a map it made during setup. At first, everything looked fine. But after a few days, issues appeared. The main problem was how people spoke to the robot. Each family member had a different way of saying things, and it's not about accents, but about the phrasing. For example, one might say, "Turn off the lamp," while someone else would ask, "Can you switch the light off?" The robot, unfortunately, didn't always know what to do. Sometimes it ignored the request altogether, and other times it would misinterpret and do the wrong task. This unpredictability led to growing frustration, especially when people were in a hurry and they expected the robot to understand the commands without additional explanations.

How could the robot be improved to work better in a home where people speak in different ways or mix languages?

LISTENING

Task 39. LISTEN FOR DETAILS. Scan the QR code and listen to people talking about their robots. Complete the table.

	Emily	**Jake**	**Anna**
1. What type of robot do they use?			
2. What do they use the robot for?			
3. What do they like about the robot?			
4. What do they find challenging about the robot?			
5. What's one unique feature of the robot?			

WATCHING

Task 40. COMMUNICATE. Work in a small group. Discuss the following questions before watching the video about space exploration robots.

1. What do you think robots do in space?
2. Why do we send robots instead of humans to other planets?
3. What challenges might space robots face?

Task 41. EXPLORE THE WORDS. Read the sentences with bold words. Then, complete each definition with the correct form of one of the words.

1. The first rover on Mars **ran out of juice** after 85 days and stopped working.
2. Mars used to have **plentiful** water on its surface.
3. Some chemicals found on Mars might be a **precursor** to life.

4. The rover sent back a **boatload** of amazing photos.
5. Helper bots are a **subset** of space robots designed to assist astronauts.
6. The robotic arm moves heavy **payloads** on the space station.
7. The Canadarm2 can **reposition** itself by moving like an inchworm.
8. The robot's **propulsion** system allows it to move in microgravity.

a) __________ (n) a group within a larger category
b) __________ (v) move something to a new position
c) __________ (adj) existing in large amounts; abundant
d) __________ (n) an early stage or ingredient that leads to something later
e) __________ (n) a large amount of something
f) __________ (n) the force that moves an object forward, often using jets or engines
g) __________ (v phrase) lose power or energy and stop working
h) __________ (n) a load carried by a spacecraft, such as cargo or equipment

Task 42. WATCH FOR MAIN IDEAS. Scan the QR code and watch the video "Space Exploration Robots". Match each robot with its purpose.

Robot Name	Purpose
1. Sojourner	a) searched for conditions suitable for life
2. Spirit & Opportunity	b) found signs of past water on Mars
3. Curiosity	c) moved cargo and assisted in docking spacecraft
4. Perseverance	d) looked for direct signs of life on Mars
5. Canadarm2	e) was the first small rover on Mars
6. CIMON	f) helped astronauts with tasks inside the ISS

Task 43. WATCH FOR DETAILS. Complete the summary with the correct words from the box.

life ▪ exploration ▪ automated ▪ rovers ▪ helper ▪ mission ▪ payloads ▪ evidence ▪

There are three main types of space robots: 1) ______ bots, which explore places humans cannot; 2) ______ bots, which assist astronauts; and 3) ______ systems, which work without human control. NASA has sent several 4) ______ to Mars, including Sojourner, which was the first to land in 1997. The Spirit and Opportunity 5) __________ found 6) ____________ of water on Mars. More advanced rovers, like Curiosity, continue to search for 7) __________. Helper bots, such as Canadarm2, assist astronauts with tasks like handling 8) ____________.

Task 44. COMMUNICATE. Work in pairs. Discuss the questions below.

1. What would you like to know about life on other planets? How could robots help us find out more?
2. Do you believe robots will one day help us live on other planets? Why or why not?
3. Do you think robots should continue to be used for space exploration, or should we prioritize sending humans to explore space?

4. What would you miss the most if robots started doing most of the work in space exploration?
5. If you had the chance to travel to space, would you prefer to go with a robot assistant or without? Why?
6. Would you like to see more robots sent to explore the Moon or Mars? What do you think they might find there?

LANGUAGE FOCUS

Task 45. STUDY AND ANALYZE. Look at the rule about reported speech in questions, study in what situations it is used.

REPORTED SPEECH IN QUESTIONS

Direct Speech	Reported Speech
Questions beginning with have, do, or be	
‘Have you programmed the robot to detect errors?’ she asked.	She asked **if I had programmed** the robot to detect errors.
‘Do you use robotic arms in your factory?’ the manager asked.	The manager asked **if we used** robotic arms in our factory.
‘Is the surgical robot working properly?’ the doctor asked.	The doctor asked **if the surgical robot was working** properly.
Questions beginning with a modal	
‘Can this robot assist in complex surgeries?’ the reporter asked.	The reporter asked **if the robot could assist** in complex surgeries.
‘Will the AI-powered robot replace human workers?’ the journalist asked.	The journalist asked **if the AI-powered robot would replace** human workers.
‘Should we invest in autonomous delivery robots?’ the CEO asked.	The CEO asked **if they should invest** in autonomous delivery robots.
‘May I test the robotic exoskeleton?’ the engineer asked.	The engineer asked **if he might test** the robotic exoskeleton.
Questions beginning with a question word	
‘What kind of sensors do industrial robots use?’ the student asked.	The student asked **what kind of sensors industrial robots used**.
‘Who developed the first humanoid robot?’ she asked.	She asked **who had developed** the first humanoid robot.
‘Which robot is the most efficient in warehouses?’ the supervisor asked.	The supervisor asked **which robot was** the most efficient in warehouses.
‘When will domestic robots be widely used?’ he asked.	He asked **when domestic robots would be widely used**.
‘Why do hospitals use robotic assistants?’ the visitor asked.	The visitor asked **why hospitals used** robotic assistants.
‘How much does a robotic pet cost?’ the customer asked.	The customer asked **how much a robotic pet cost**.

Task 46. PRACTICE. Circle the correct answer.

1. ‘Has the robot finished the task?’ the supervisor asked.
   The supervisor asked if the robot **has finished / had finished** the task.
2. ‘Do robots work faster than humans?’ the manager asked.
   The manager asked if robots **work / worked** faster than humans.
3. ‘Is this device new?’ the technician asked.

The technician asked if the device **is / was** new.
4. 'Have they repaired the equipment?' the engineer asked.
The engineer asked if they **have repaired / had repaired** the equipment.
5. 'Does the system check for errors?' the inspector asked.
The inspector asked if the system **checked /checks** for errors.
6. 'Are robots helping in factories?' the journalist asked.
The journalist asked if robots **had been helping / were helping** in factories.

Task 47. PRACTICE. Write one word in each gap.

1. 'Will the system finish the task?' the engineer asked.
The engineer asked if the system __________ finish the task.
2. 'May I test the assembly line?" the technician asked.
The technician asked if he __________ test the assembly line.
3. 'Can we increase the robot's speed?' the production manager asked.
The production manager asked if they __________ increase the robot's speed.
4. 'Shall we add a camera to the robot?' the engineer asked.
The engineer asked if they __________ add a camera to the robot.
5. 'May I control it remotely?' the worker asked.
The worker asked if he __________ control it remotely.
6. 'Will this solution reduce costs?' the boss asked.
The boss asked if this solution __________ reduce costs.

Task 48. PRACTICE. Rewrite as reported questions, beginning with the words given.

1. 'Has the vacuum finished cleaning the living room?'
My mum asked me if __________.
2. 'Can I schedule the irrigation system for early morning?'
The farmer wanted to know whether __________.
3. 'Are you still using the robot to monitor patients?'
The doctor asked if __________.
4. 'Does the machine have to scan the crops every day?'
I asked the engineer if __________.
5. 'Do you want the assistant to prepare your meal?'
Jan asked me if __________.
6. 'Can you guess what settings I used for the robotic surgery?'
The surgeon asked Wendy if __________.
7. 'Do you trust the automated caregiver?'
Fred asked Gloria whether __________.
8. 'Will the drone be delivering the pesticides tomorrow?'
Adrian wondered whether __________.

Task 49. PRACTICE. Rewrite as direct questions.

1. She asked me why the home assistant had stopped responding.
2. He asked her what the difference was between a robotic milking system and a traditional one.
3. They asked us how the autonomous tractor had performed the day before.
4. I asked them when they had last used the robotic exoskeleton for rehabilitation.
5. She asked him which type of robotic vacuum he preferred.
6. I asked you how you were going to program the drone for field monitoring.
7. Carl asked Megan who had operated the robotic surgical arm the previous week.
8. Megan asked Carl what gave him the right to decide when the house-cleaning assistant should run.

UNIT 1.3 ROBOTICS AND SOCIETY

READING AND VOCABULARY

Task 50. READ FOR DETAILS. Read the following short news stories about robotics. Some of them are real, others are fictional. After reading, decide whether each article seems true or false. Discuss your reasoning based on logic, known trends in robotics, and examples of how these technologies might work in real life.

RoboLawyer Wins Landmark Court Case

There's been a lot of talk online about a supposed legal breakthrough. A company called DoNotPay has developed an AI tool called *RoboLawyer* that, according to reports, helped win a high-profile criminal case. The AI system could synthesize legal arguments in real time and even respond using voice synthesis technology. While some experts see this as a major step forward, others question whether such tech is truly ready for courtroom use.

Self-Healing Robots Repair Themselves

XBotics, a robotics company known for its cutting-edge innovations, says it has created a new kind of self-healing robot. These machines are built with special soft materials that contain polymers which bond together when exposed to heat or pressure. The company's *XHealBot* model is already in testing and appears to show improved durability and an extended lifespan which could make them ideal for use in rough environments or situations where repairs are hard to manage.

AI Nanny Cares for Children Without Human Supervision

TechGuard recently introduced a robot called the *Guardian Nanny*, designed to handle childcare with no human intervention at all. It can cook, clean, teach, and monitor children using AI decision-making systems. The robot's ability to analyze behavior and respond to children's needs has raised eyebrows. Although it's being promoted as a 24/7 solution, many are skeptical about the idea of replacing human caregivers entirely, especially in something as sensitive as child safety.

Underwater Robots Discover a New Species

Ocean Infinity has reported that one of its underwater robots, *SeaBot*, helped identify a new bioluminescent species during a recent mission. The robot explored hard-to-reach areas of the ocean, deep-sea trenches and used advanced sensors and AI tools to record images and collect real-time data. Scientists hope that this discovery will add to our understanding of marine biodiversity, and believe that robots like SeaBot can open doors to future discoveries in places humans can't easily go.

Humanoid Robots Now Eligible for Citizenship

A few years ago, the humanoid robot *Sophia*, developed by Hanson Robotics, was officially granted citizenship by Saudi Arabia. The announcement with this historic decision made headlines worldwide and sparked debate over the rights of robots in society. Sophia, known for her human-like appearance and ability to hold conversations, became the first robot to receive such a status and legal recognition. Whether this move was mostly symbolic or a sign of things to come is still being discussed.

Robot Researchers Now Conduct Full-Scale Scientific Studies

IBM recently presented a groundbreaking development, a project called the *Robotic Research Assistant* (RRA), where a robot can supposedly run full scientific investigations. According to IBM, the system can generate hypotheses, conduct experiments, analyze results, and even write up and publish papers. The idea sounds ambitious – a machine carrying out research from start to finish.

And while some call it revolutionary, others believe it's still a long way from replacing human researchers.

Spot the Robot Joins Search-and-Rescue Missions

After a recent earthquake in Japan, Boston Dynamics' robot *Spot* was used to assist teams in search-and-rescue missions. The four-legged robot is equipped with sensors and cameras that allow it to move across rough terrain and send back valuable information. Its ability to navigate dangerous zones has made it especially helpful in locating survivors where people or drones might not be able to reach safely.

- What clues helped you understand whether a story was true or false?
- How do media influence public perception of robotics?
- What ethical concerns arise from the spread of fake news about AI and robotics?

Task 51. WORK WITH WORDS. Find the words in the articles above based on these definitions.

1. _______________ (n) the ability to last a long time without being damaged.
2. _______________ (v) to combine different elements into a whole.
3. _______________ (adj) difficult to access or get to.
4. _______________ (n) significant and important discovery or development.
5. _______________ (n) the surface features of an area.
6. _______________ (n) the length of time something lasts or remains functional.
7. _______________ (n) ideas that can be tested through research or experiments.
8. _______________ (adj) the most advanced and modern technology or development.
9. _______________ (adj) producing light naturally by organisms.
10. _______________ (n) the legal status of being a recognized member of a country.
11. _______________ (adj) being able to repair itself without external help.
12. _______________ (n) the variety of living organisms in a particular environment.

Task 52. WORK WITH WORDS. Match the two parts of collocations from the articles.

1. **search-and-rescue**	a) decision
2. **voice synthesis**	b) missions
3. **groundbreaking**	c) appearance
4. **historic**	d) recognition
5. **real-time**	e) experiments
6. **legal**	f) intervention
7. **deep-sea**	g) data
8. **human**	h) development
9. **conduct**	i) trenches
10. **human-like**	j) technology

Task 53. WORK WITH WORDS. Complete the sentences by changing the form of the word in brackets to fit the context.

1. The robot's ability to act without human control makes it highly ___________. (autonomy)
2. Engineers have developed a new AI system that can ___________ human speech with incredible accuracy. (synthetic)
3. Sophia, the humanoid robot, was the first to receive official ___________ as a citizen. (recognize)
4. The recent ___________ of an AI capable of creative thinking has amazed scientists. (discover)

5. Many people believe that AI will never fully replace human __________ in complex tasks. (assist)
6. The research team performed a detailed __________ of the robot's performance in extreme conditions. (analyze)
7. The company made a historic __________ to grant robots limited legal rights. (decide)
8. The development of self-healing robots is considered a __________ step in robotics. (revolution)
9. The new robotic nanny has been criticized due to concerns about the __________ of leaving children under AI supervision. (safe)
10. Boston Dynamics has designed a robot that is highly __________ and can adapt to rough terrain. (durability)

SPEAKING

Task 54. COMMUNICATE. Work in pairs. Study the three laws of robotics introduced by a science fiction writer Isaac Asimov. Discuss the questions below.

1. Why do you think Asimov created these laws?
2. Can these laws work in real life? Why or why not?
3. Imagine a robot is in a situation where it must choose between two people in danger. What should it do?
4. Should robots always follow human orders? Why or why not?
5. What happens if a robot ignores the Third Law (self-preservation)?
6. Think about robots in healthcare, military, or customer service. Do these laws protect people in these areas?
7. Would you add a Fourth Law to improve robot behavior? What would it be?
8. Some people believe AI can be dangerous. How should these laws change for AI systems?
9. In the future, should robots be allowed to make decisions instead of humans (e.g., in law enforcement or medicine)? Why or why not?
10. Who is responsible if a robot makes a bad decision – the robot, the programmer, or the company?

Task 55. COMMUNICATE. Work in pairs. Discuss the robot dilemmas below.
Robots are designed to follow Asimov's Three Laws of Robotics, but sometimes they face difficult situations where the laws may not work perfectly. Discuss each scenario with your partner and decide how the robot should act. Be ready to explain your decision to the class.

- A hospital robot receives two emergency calls at the same time. One patient is critically ill but far away, and another patient is less ill but nearby. Who should it help first?
- A robot police officer is ordered to arrest an innocent person. Should it follow the order, even if it conflicts with Asimov's First Law?
- A factory robot detects a fire in the building. If it tries to put out the fire, it may get damaged beyond repair. If it doesn't, workers might be at risk. What should it do?
- A robot judge in a court case has all the legal evidence to make a decision, but it doesn't understand human emotions and intentions. Should robots be allowed to make legal decisions?

Task 56. COMMUNICATE. Work in pairs. Act out the situations below. One of you is a human, and the other is a robot. The human gives commands, but the robot must respond following Asimov's Three Laws of Robotics. After each conversation, the class will decide: Did the robot follow the laws correctly? Could the robot have answered differently?

1. Human: "Robot, open the door!"
   Problem: A dangerous person is outside. Should the robot obey?
2. Human: "Robot, bring me my medicine!"
   Problem: The medicine is not safe for the human. What should the robot do?
3. Human: "Robot, protect yourself!"
   Problem: If the robot protects itself, a human might get hurt. What should it do?
4. Human: "Robot, get me out of the burning building!"
   Problem: The robot can save only one – you or itself. What choice should it make?
5. Human: "Robot, turn off all security cameras!"
   Problem: The cameras protect people in the building. Should the robot obey?
6. Human: "Robot, save me first!"
   Problem: There are two other injured people. Who should the robot help first?
7. Human: "Robot, drive me home as fast as possible!"
   Problem: The robot driver must choose between running a red light (risking an accident) or obeying traffic rules but delaying the human. What should it do?
8. Human: "Robot, take money from that store to help my starving family!"
   Problem: The robot should not steal, but the family needs food to survive. How should it respond?

Task 57. COLLABORATE. Work in small groups. Read each dilemma and discuss possible solutions.

Elderly Care Robots

- A robot detects a medical emergency but cannot reach a doctor. Should it act on its own?
- An elderly person refuses to use a care robot. Should their family insist?
- A malfunction causes the robot to give the wrong medication. Who is responsible?
- A robot taking care of an elderly person notices they want to eat unhealthy food that could be dangerous for their health. Should the robot allow it or stop them?

Medical Robots

- A hospital AI finds a mistake in a doctor's diagnosis but is not 100% sure. Should it tell the patient?

- A surgery robot breaks down in the middle of an operation. Should it try to finish or stop completely?
- A mental health chatbot gives advice to people with problems, but it cannot call for help in an emergency. Should AI be used for serious mental health support?

Work & Factory Robots

- A factory robot replaces workers to do the job faster. Should the company help workers find new jobs?
- A building robot finds a safety problem, but stopping work will cost a lot of money. Should it stop or continue?
- A robot in a warehouse notices an employee taking items without permission. Should it report the person immediately or wait for a human to check?

Security Robots

- A security robot sees someone in a restricted area but doesn't know if they are dangerous. Should it act or wait for a human?
- A security AI scans people at an airport and flags someone as a threat. There is no proof, but the AI is 80% sure. Should security arrest the person?
- A police robot is patrolling a neighborhood when it detects a crime. Should it try to stop the criminal or just call for human officers?

Task 58. COLLABORATE. Divide into two teams. The teams should take turns reading the facts and deciding if they are True or False, then they bet on their answer (10-100 points). The referee or teacher says the answer. If their answer is correct the bet is added to the total, if not, it is deducted. Each team starts with 1 000 points. The team with the highest score at the end wins.

Team A			**Team B**		
Question	**Bet**	**New Total 1 000 points**	**Question**	**Bet**	**New Total 1 000 points**
1. e.g. Correct	20	+20 (1020)	1.		
2.			2.		
3.			3.		
4.			4.		
5.			5.		
6.			6.		
7.			7.		
8.			8.		
9.			9.		
10.			10.		

1. Leonardo da Vinci drew up plans for an armored humanoid machine in 1495. Engineer Mark Rosheim reconstructed his designs and created a functional miniature for NASA to help colonize Mars.
2. More than a million industrial robots are now in use, nearly half of them in Japan.
3. The first known case of robot murder occurred in 1981, when a robotic arm crushed a Japanese Kawasaki factory worker.
4. Twin robots Spirit and Opportunity were designed for 90-day mission but functioned for 11 years.

5. Taliban fighters in Afghanistan have reportedly used ladders to flip over and disable the U.S. military robots sent to scout out their caves.
6. Elektro, the world's first humanoid robot, debuted in1939. Built by Westinghouse, the seven-foot-tall walking machine "spoke" more than 700 words stored on 78-rpm records to simulate conversation.
7. A robot called Raptor runs three times as fast as a human.
8. R2-D2 is the only character that appears unchanged in all six Star Wars movies.
9. Chris Melhuish of the Bristol Robotics Laboratory created robots that use bacteria-filled fuel cells to produce electricity from rotten apples and dead flies. The goal: robots that forage for their own food.
10. The largest flying robot insect, the BionicOpter, has its origins in studies of the flight dynamics of bees.
11. Sophia, a social humanoid robot, is the first robot to get citizenship.
12. Most computerized robot voices tend to be male.
13. Robear, looking similar to a bear, is a military robot designed to reduce the number of soldiers in the army. It's capable of attacking and transporting heavy weights.
14. The first robot policeman appeared in Perm, Russia.
15. Curiosity was the latest launched robot on Mars.
16. Famous people who suffered from robophobia (an irrational fear of robots) include Stephen Hawking, Elon Musk, Jaron Lanier, Bill Gates, and Kiibo.
17. Cybernetics professor Kevin Warwick calls himself the world's first cyborg, with computer chips implanted in his left arm. He can remotely operate doors, an artificial hand, and an electronic wheelchair.
18. Winebot, built by Japan's NEC System Technologies and Mie University, can recognize different wines, cheeses, and refreshments . . . up to a point. It recently mistook a reporter's hand for prosciutto.
19. The terms "android" and "robot" mean the same.
20. Researchers at the University of Stanford created a machine learning algorithm that is capable of predicting death with a shocking 98 % accuracy.

LISTENING

Task 59. COMMUNICATE. You are going to listen to a podcast about how movies and books have influenced robotics. Before listening, discuss with your partner:

- Can you name a movie or book that influenced real-life robots?
- Do movies show robots in a realistic way? Why or why not?
- How do films and books change the way people feel about robots?

Task 60. LISTEN FOR DETAILS. Scan the QR code and listen to the podcast where David Carter interviews Dr. James Reynolds about robots in movies and books. Then, answer the questions below.

1. Why are Isaac Asimov's Three Laws of Robotics important?
2. How did early movies like Metropolis influence robot design?
3. Which movie makes robots seem dangerous?
4. How does real robotics help filmmakers today?
5. How do robots in films inspire real engineers?
6. Why does the guest believe that robotics and science fiction create a cycle of inspiration?

Task 61. COMMUNICATE. Work in pairs. Choose a famous robot from movies or literature and analyze how realistic it is. Use the internet if needed. Then, present your findings to the class.

Discuss the following questions:

1. How realistic is this robot? Could a similar one exist today?
2. What real technologies (AI, sensors, robotics, etc.) would be needed to build it?
3. How does this robot compare to actual robots used in the real world?

Choose a robot from the list below or select one of your own:

- R2D2 (Star Wars)
- C3PO (Star Wars)
- The Terminator (The Terminator)
- WALL-E (WALL-E)
- Ava (Ex Machina)
- Optimus Prime (Transformers)
- Data (Star Trek)
- Baymax (Big Hero 6)
- Sonny (I, Robot)
- Bender (Futurama)
- TARS (Interstellar)
- Chappie (Chappie)

READING AND VOCABULARY

Task 62. READ FOR MAIN IDEA. Read the article quickly to understand the general idea. Which of the following topics is NOT mentioned?

A. AI companionship as a solution to loneliness.
B. The technological features of AI companions.
C. The use of AI companions in professional workplaces.
D. The ethical concerns of AI relationships.
E. The effect of AI companions on mental health.

Task 63. READ FOR DETAILS. These sentences have been removed from the article. Read the article again and match each sentence with a numbered gap.

A. Ethical issues also arise.
B. One of the most important is the loneliness pandemic.
C. This happens thanks to factors such as memory storage and empathy simulation.
D. Replika is just one of the known AI chatbots that learns from text messages to form emotional connections.
E. Others see it as a potential replacement for human relationships in general.
F. This has a positive effect on people with social anxiety.

AI Companions and Human Relationships

The concept of AI companions is not new, but has only recently been put to practical use. Such AI companions are emotional assistants, and in the case of Gatebox, they mimic romantic relationships. 1) ___________________ Gatebox takes it a step further by creating a holographic AI avatar that sends messages, greets the user at home, and can control smart homes. Harmony from RealDoll combines artificial intelligence with a humanoid robot to provide communication and companionship.

People choose AI companions for a number of reasons. 2) ______________________ In countries like Japan, where social life is individualistic, AI companions provide a sense of belonging. They are also available 24/7, don't have emotional baggage and are always ready to respond. Advances in AI are making such interactions increasingly human-like. 3) ________________

While AI companions have their advantages, they also have disadvantages. Some argue that it can help people become socially confident and improve emotional well-being. AI companions provide a space where a person can behave freely without being criticized. 4) ______________ However, there are concerns that reliance on AI can increase social isolation, making human relationships less appealing. 5) ________________ AI never feels emotion, but people become emotionally close to it, which calls into question the concepts of empathy and love.

As AI companions improve, the impact on relationships is inevitable. Some see AI as an enhancement to human interaction, a useful social practice. 6) _________________ The reproducibility and the customizable nature of AI are attractive traits, especially for those who have difficulty with human relationships. This new phenomenon is challenging traditional views of love and friendship, which forces society to reconsider what it means to connect with others.

(adapted from https://www.forbes.com/sites/neilsahota/2024/07/18/how-ai-companions-are-redefining-human-relationships-in-the-digital-age/)

Task 64. WORK WITH WORDS. Match each highlighted word from the article with its meaning.

1. **empathy**	a) to imitate someone or something.
2. **replacement**	b) the ability to understand and share another person's feelings.
3. **anxiety**	c) a state of being healthy, happy, and comfortable.
4. **mimic**	d) the state of depending on something or someone.
5. **companionship**	e) the ability to be copied or repeated exactly.
6. **well-being**	f) attractive or interesting.
7. **reliance**	g) the feeling of friendship and being with someone.
8. **appealing**	h) certain to happen and impossible to avoid.
9. **inevitable**	i) something that takes the place of another.
10. **reproducibility**	j) a feeling of worry or nervousness.

Task 65. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Can robots or AI companions truly provide emotional support, or is it just an illusion of companionship?
2. How does robot companionship compare to human relationships in terms of emotional depth?
3. Do you think robot companions can help reduce loneliness, or could they make social isolation worse?
4. How might the increasing use of domestic robots affect traditional friendships and romantic relationships?
5. Is it ethical to form emotional bonds with a robot that cannot truly feel emotions?
6. Should there be regulations on domestic robots to prevent emotional dependence?

LISTENING

Task 66. COMMUNICATE. Work in pairs. Read the description of the sci-fi TV series and answer the questions below.

> ***Westworld*** is a sci-fi TV series set in a futuristic theme park where guests can live out their wildest fantasies. The park is filled with AI robots, called "hosts," who look and act like real humans. The guests can do whatever they want in the park, and the hosts must follow their programmed roles.

1. What problems do you think could happen in a world where AI robots look and act like humans?
2. Do you think a theme park like *Westworld* would be fun or dangerous? Why?

Task 67. LISTEN FOR DETAILS. Scan the QR code and listen to a podcast “AI, Robots, and the Future – A Look at Westworld”. Then read the statements and choose the correct answer.

1. What is Westworld about?
   A. A world where AI controls humans
   B. A futuristic theme park with AI robots
   C. A city where people live with advanced technology
2. What happens to some AI hosts in Westworld?
   A. They start to remember past experiences and become self-aware
   B. They break down and stop functioning
   C. They begin acting exactly like humans and take over the world
3. Why do some guests in Westworld form emotional connections with the hosts?
   A. The hosts have emotions like humans
   B. The guests don’t know the hosts are robots
   C. The hosts never judge, argue, or refuse to listen
4. What real-world connection does the podcast mention?
   A. AI companions like chatbots and humanoid robots are becoming more common
   B. Some people already live with AI robots in their homes
   C. AI is replacing human relationships completely
5. What is one danger mentioned about AI in the future?
   A. AI will never be able to think for itself
   B. AI will always be completely controlled by humans
   C. AI might become too smart and decide it doesn’t need humans
6. What is the main message of the podcast?
   A. AI companions are better than human relationships
   B. AI is useful but raises ethical questions we need to consider
   C. AI will take over the world soon

Task 68. COMMUNICATE. Work in pairs. Discuss the questions below.

1. The series explores the idea of free will. Do the hosts make real choices, or are they always following their programming?
2. Many guests in Westworld treat the hosts cruelly because they aren’t "real." Do you think people’s actions in a virtual world reflect their true nature?
3. The series suggests that AI can predict and control human behavior. Do you think this is realistic? Could AI be used to manipulate people in real life?
4. If robots become very advanced, should they have rights like humans? Why or why not?
5. If robots could feel pain, should we treat them like living things?

Task 69. COMMUNICATE. Work in groups. Read the film descriptions and discuss the questions.

Her (2013)

A lonely writer falls in love with his AI assistant, Samantha, who has no physical form but a highly advanced personality.

- Could you fall in love with an AI that has no body?
- What are the benefits and dangers of such a relationship?
- How does Samantha’s lack of a physical

Ex Machina (2014)

A young programmer is invited to test a highly advanced humanoid AI, Ava, who appears to have consciousness – but does she really?

- How can we understand if an AI is truly conscious or just mimicking emotions?
- Would you trust a robot that behaves like a human? Why or why not?
- If a highly advanced AI wanted to escape

form affect the relationship? • If an AI can provide emotional support and companionship, should it be considered a life partner?	human control, would that mean it has true consciousness?
Humans (TV Series, 2015-2018) ***In an alternative present, humanoid robots called "Synths" are used as household assistants. But some Synths develop emotions and self-awareness.*** • Would you want a humanoid robot as a personal assistant? Why or why not? • What risks could arise from robots being too human-like? • How might society change if robots with emotions became common? • If a robot looks, speaks, and acts like a human, what makes it different from us?	**Black Mirror: "Be Right Back" (2013, S2E1)** ***A woman orders an AI version of her dead boyfriend, built from his online data. At first, he seems real, but she soon realizes he lacks something essential.*** • Would you want an AI copy of a loved one who passed away? • Can AI ever replace real human memories and relationships? • Should technology allow us to bring back the dead in digital form? • If an AI can learn from a person's past messages, does that make it them? Why or why not?

Choose one film/series you haven't seen. Based on the description, predict how the story will end.

WRITING

Task 70. ANALYZE. Read the letter below from an IT Manager to a robotics company. Then match each paragraph with the correct heading from the list. After that, identify what specific details the writer is asking about.

A. Opening remarks / reason for writing
B. Specific questions about product features
C. Request for further communication
D. Closing remarks
E. Description of current needs

Dear Ms. Carter,	
My name is Elena Flint, and I am the IT Manager at NextGen Solutions. I am writing to inquire about advanced robotic arms for our new automated assembly line.	Para 1: ____________
Our company is currently upgrading its manufacturing process and needs robotic arms that can handle precision tasks, such as soldering, component placement, and fine motor assembly. The robots must be compatible with our existing conveyor system and work safely alongside human operators.	Para 2: ____________
Could you provide detailed information about the available models that match these needs? We are especially interested in features like load capacity, motion range, accuracy, and ease of programming. Please also include pricing for bulk purchases, warranty terms, and expected delivery times.	Para 3: ____________
We would appreciate the opportunity to schedule a call or online meeting to discuss possible cooperation and custom configuration options.	Para 4: ____________
Thank you for your attention. I look forward to hearing from you soon.	Para 5: ____________

Yours sincerely,
Elena Flint
IT Manager
NextGen Solutions

Task 71. WRITE. You are the IT Manager at RusAutomation. Your company is looking to implement robots in a warehouse environment to help with package sorting and logistics. Write a formal letter of inquiry to Robotics & Motion Systems Ltd. Ask for detailed information about suitable robots for your task. You may use the ideas below or add your own.

- State who you are and why you are writing.
- Describe the task the robot should perform (e.g., package sorting, barcode scanning, stacking).
- Ask about important features (speed, accuracy, load capacity, integration with existing software).
- Ask about pricing, support, and customization options.
- End with a polite closing and request for a reply or meeting.

Useful phrases:

- I am writing to inquire about...
- Our company requires robots that can...
- We are interested in models that offer...
- Could you also include details about...
- I would appreciate the opportunity to discuss...
- I look forward to your response.

LANGUAGE FOCUS

Task 72. STUDY AND ANALYZE. Look at the rule about reported speech in commands, suggestions and requests, study in what situations they are used.

REPORTED SPEECH IN COMMANDS, SUGGESTIONS AND REQUESTS

	Direct Speech	Reported Speech
+ to Infinitive **agree**	‘Yes, I’ll repair the robot.’	The engineer **agreed to repair** the robot.
claim	‘I have tested the robotic surgeon successfully.’	The doctor **claimed to have tested** the robotic surgeon successfully.
demand	‘Check the robotic assistant’s battery level immediately!’	The supervisor **demanded to check** the robotic assistant’s battery level immediately.
offer	‘Would you like me to install the new AI chatbot?’	The technician **offered to install** the new AI chatbot.
promise	‘I will have calibrated the robotic arm by tomorrow.’	The scientist **promised to calibrate** the robotic arm by the next day.
refuse	‘No, I won’t allow the cleaning robot to work at night.’	The supervisor **refused to allow** the cleaning robot to work at night.
threaten	‘If you don’t fix the robot, I’ll stop using it.’	The doctor **threatened to stop** using the robot if it wasn’t fixed.
+ somebody+to Infinitive **advise**	‘You should check the robot’s sensors	The engineer **advised the team to**

	regularly.'	**check** the robot's sensors regularly.
allow	'You can use the robot to assist in the operation.'	The surgeon **allowed the team to use** the robot to assist in the operation.
ask	'Can you check if the space rover is fully charged?'	The scientist **asked the engineer to check** if the space rover was fully charged.
beg	'Please teach me how to control the rescue robot!'	The trainee **begged the instructor to teach** him how to control the rescue robot.
command	'Deploy the rescue robot immediately!'	The officer **commanded the team to deploy** the rescue robot immediately.
forbid	'Do not let the delivery robot enter the restricted zone!'	The supervisor **forbade the workers to let** the delivery robot enter the restricted zone.
invite	'Come and see how the medical robot performs surgery!'	The doctor **invited the students to see** how the medical robot performs surgery.
order	'Turn off the AI assistant now!'	The supervisor **ordered the employee to turn off** the AI assistant.
remind	'Don't forget to test the robot's sensors before the flight.'	The engineer **reminded the astronaut to test** the robot's sensors before the flight.
warn	'Be careful! The robot might move suddenly.'	The technician **warned the workers to be** careful because the robot might move suddenly.
+-ing **accuse of**	'You damaged the robot's sensor!'	The technician **accused him of damaging** the robot's sensor.
admit to	'I made a mistake in programming the robot.'	The engineer **admitted to making** a mistake in programming the robot.
apologize for	'I'm sorry for shutting down the robotic assistant by mistake.'	The technician **apologized for shutting down** the robotic assistant by mistake.
boast about/of	'Our AI system completed the surgery faster than any human doctor!'	The hospital director **boasted about the AI system completing** the surgery faster than any human doctor.
complain to smb of	'The robot vacuum isn't working properly!'	The customer **complained to the store of the robot vacuum not working** properly.
deny	'I didn't ignore the security robot's warning.'	The worker **denied ignoring** the security robot's warning.
insist on	'We must use the robotic arm for this delicate procedure.'	The surgeon **insisted on using** the robotic arm for the delicate procedure.
suggest	'Why don't we program the robot to assist with customer calls.'	The developer **suggested programming** the robot to assist with customer calls.
+that clause **explain**	'The robot stopped working because of a software bug.'	The engineer **explained that the robot had stopped working** because of a

		software bug.
inform smb	'The delivery robot will arrive at 3 p.m.'	The technician **informed the customer that the delivery robot would arrive** at 3 p.m.
exclaim/remark	'This robot can solve complex equations in seconds!'	The scientist **exclaimed that the robot could solve** complex equations in seconds.

Task 73. PRACTICE. Complete the sentences.

1. 'You damaged the eldercare robot's navigation system!'
   The supervisor accused the trainee of …
2. 'The robot stopped working because of a battery failure.'
   The technician explained that **...**
3. 'This robotic dog doesn't follow commands properly!'
   The customer complained to the store of ...
4. 'Can you check if the domestic humanoid robot has finished cleaning?'
   The homeowner asked ...
5. 'You are not allowed to use the robotic assistant without supervision!'
   The manager forbade ...
6. Don't forget to update the care robot's software."
   The developer reminded ...
7. 'I'm sorry for disconnecting the domestic robot from Wi-Fi.'
   The technician apologized for ...
8. 'This household robot can cook an entire meal in just 10 minutes!'
   The scientist exclaimed that ...
9. 'Why don't we install a voice assistant in the robot?'
   The designer suggested ...
10. 'Our smart home system is the most advanced on the market!'
    The company representative boasted about ...

Task 74. PRACTICE. Complete the sentences with the correct form of a reporting verb from the box.

deny ▪ insist ▪ advise ▪ suggest ▪ accuse ▪ threaten ▪ boast ▪ promise ▪ warn ▪ agree ▪ complain ▪ remind

1. 'If you don't follow the safety rules, I'll deactivate the cobot!'
   The manager ______ to deactivate the cobot if the safety rules weren't followed.
2. 'The robotic toy is making too much noise!'
   The parent ______________ about the robotic toy making too much noise.
3. 'Why don't we program the space rover to collect more samples?'
   The scientist ___________ programming the space rover to collect more samples.
4. 'Ok. I will help you install the new AI assistant in your cobot.'
   The technician ___________ to help install the new AI assistant in the cobot.
5. 'I didn't cause the robotic arm to malfunction!'
   The technician ____________causing the robotic arm to malfunction.
6. 'You should let the cobot handle the heavy lifting.'
   The engineer ___________ using the cobot for heavy lifting.
7. 'I will make sure the robotic toy gets new interactive features.'
   The developer ___________ to add new interactive features to the robotic toy.

8. ‘Be careful! The robotic dog might knock over small objects!’
   The pet store employee ______ the customer about the robotic dog knocking over small objects.
9. ‘You changed the cobot’s programming without permission!’
   The manager ______ the intern of changing the cobot’s programming without permission.
10. ‘Always recharge the robotic guide dog before taking it outside.’
    The trainer ______ his client to recharge the robotic guide dog before going outside.
11. ‘I am certain the robotic firefighter didn’t fail to detect heat.’
    The emergency response team ______ on the robotic firefighter detecting heat correctly.
12. ‘Our robotic delivery service is the fastest in the industry!’
    The company ______ about their robotic delivery service being the fastest in the industry.

Task 75. PRACTICE. Turn the following sentences from direct into reported speech and vice versa.

1. ‘The space rover isn’t sending back clear images, complained the scientist.
2. The surgeon insisted on using AI-powered surgical robots to reduce human error.
3. The AI specialist suggested adding facial recognition to the security robot.
4. ‘The robotic nurse should give medication at exact intervals,’ explained the doctor.
5. ‘We will integrate an emergency stop function into the cobot,’ said the team.
6. The customer boasted about having the latest version of the robotic chef.
7. ‘Be careful! The drone might lose connection in strong winds!’ said the scientist.
8. The astronaut insisted on testing the robot’s durability before the space mission.